U S T E T 2 0 1 4

MATHEMATICS PROFICIENCY
Time -- 50 minutes
60 Questions

NAME: _____
SCHOOL: _____

For each question, decide which is the best answer choice, and darken the corresponding oval on the answer sheet.

1. The product of two numbers is -108. Their sum is 3. Find the larger of the two numbers.
   A. 9
   B. 12
   C. 15
   D. 18

2. The sum of two numbers is 13. One number is seven less than thrice the second number. Find the larger of the two numbers.
   A. 7
   B. 8
   C. 5
   D. 12

3. Two numbers differ by 5. The larger is two less than twice the smaller. Find the smaller of the two numbers.
   A. 7
   B. 12
   C. 8
   D. 2

4. Given the equation below, solve for x:
$$\sqrt{2x+2} = 3 - \sqrt{x}$$
   A. -49
   B. -1
   C. 1
   D. 49

5. Find the domain of the function $f(x) = \frac{x-3}{x+5}$.
   A. $\{x \mid x \in \Re\}$
   B. $\{x \mid x \in \Re, x \neq 5\}$
   C. $\{x \mid x \in \Re, x \neq -5\}$
   D. $\{x \mid x \in \Re, x \neq 3\}$

6. Find the domain of the function $f(x) = \frac{2x+5}{3x-4}$.
   A. $\{x \mid x \in \Re\}$
   B. $\{x \mid x \in \Re, x \neq 4/3\}$
   C. $\{x \mid x \in \Re, x \neq -5/2\}$
   D. $\{x \mid x \in \Re, x \neq 0\}$

7. Find the domain of the function $f(x) = \frac{1}{2x+1}$.
   A. $\{x \mid x \in \Re\}$
   B. $\{x \mid x \in \Re, x \neq 1/2\}$
   C. $\{x \mid x \in \Re, x \neq -1/2\}$
   D. $\{x \mid x \in \Re, x \neq 1\}$

8. Fill in the blank.
   $(2a - 3b + 1)^2 = 4a^2 + 9b^2 + 1 +$ ____ $+ 4a - 6b$
   A. $12ab$
   B. $6ab$
   C. $-6ab$
   D. $-12ab$

9. Simplify: $(\sqrt{5}+3)(\sqrt{5}-3)$
   A. 2
   B. -4
   C. $\sqrt{5}-9$
   D. 4

10. Simplify: $(4-\sqrt{10})(4+\sqrt{10})$
   A. 6
   B. -6
   C. $\sqrt{10}$
   D. 4

11. Simplify: $(\sqrt{7}-2)(\sqrt{7}+2)$
    A. 1
    B. -2
    C. $\sqrt{7}$
    D. 3

12. Simplify: $(2\sqrt{3}-5)(2\sqrt{3}+5)$
    A. -19
    B. -13
    C. 12
    D. 5

13. Solve for x: $2(x-3)+3(x+1)=4(x-2)$
    A. -5
    B. 5
    C. 6
    D. -1

14. Solve for x: $3(2x+5)-2(x-4)=3(4x+5)$
    A. -2
    B. 1
    C. 8
    D. -1

15. Factor completely: $25ax^2-49ay^2$
    A. $(5ax+7ay)(5ax-7ay)$
    B. $a(5x+7y)(5x-7y)$
    C. $a(5x+7y)(5x+7y)$
    D. $(5x+7y)(5x-7y)$

16. Factor completely: $12x^2y-15xy^2+3xy$
    A. $3xy(4x-5y+1)$
    B. $3xy(4x-5y)$
    C. $3(4x^2y-5xy^2+xy)$
    D. $3xy(4x-5y)(4x+5y)$

17. Factor completely: $x^2-12xy-45y^2$
    A. $(x-5y)(x+9y)$
    B. $(x-3y)(x+15y)$
    C. $(x-9y)(x+5y)$
    D. $(x-15y)(x+3y)$

18. Factor completely: $6x^2-19x+10$
    A. $(x-5)(6x+2)$
    B. $(3x-5)(2x-2)$
    C. $(2x-5)(3x-2)$
    D. $(6x-5)(x+2)$

19. Find the product: $(3x-5y)^2$
    A. $9x^2-30xy+25y^2$
    B. $9x^2-25y^2$
    C. $9x^2+25y^2$
    D. $9x^2+30xy+25y^2$

20. Find the product: $(7xy+5z)(7xy-5z)$
    A. $49x^2y^2-70xyz+25z^2$
    B. $49x^2y^2-25z^2$
    C. $49x^2y^2+25z^2$
    D. $49x^2y^2+70xyz+25z^2$

21. Find the product: $(2a-b-3c)^2$
    A. $4a^2+b^2+9c^2-4ab-12ac+6bc$
    B. $4a^2+b^2+9c^2$
    C. $4a^2+b^2+9c^2-2ab-6ac+3bc$
    D. $4a^2+b^2+9c^2+4ab+12ac+6bc$

22. Find the product: $(3x+7)(5x-2)$
    A. $15x^2+41x-14$
    B. $15x^2-41x-14$
    C. $15x^2-29x+14$
    D. $15x^2+29x-14$

23. Simplify: $(3xy^4z^2)^2$
    A. $9x^2y^8z^4$
    B. $9x^2y^{16}z^4$
    C. $3x^2y^8z^4$
    D. $3x^2y^{16}z^4$

24. Simplify: $(-5a^5b^6)(3ab^3)$
    A. $-15a^5b^{18}$
    B. $-15a^6b^{18}$
    C. $-15a^6b^9$
    D. $15a^6b^9$

25. Simplify: $(4xy^{-2}z^{-1})^{-2}$

A. $\frac{y^4z^2}{16x^2}$

B. $\frac{16y^4z^2}{x}$

C. $\frac{16y^4z^2}{x^2}$

D. $-\frac{y^4z^2}{16x^2}$

26. Simplify: $\frac{14a^5}{16a^3}$

A. $\frac{7a^2}{8}$

B. $-\frac{7a^2}{8}$

C. $\frac{a^2}{2}$

D. $\frac{14a^2}{16}$

27. Simplify: $(-3a^2b^5c)(14ab^3c^5)(-ab)$

A. $-42a^3b^8c^6$

B. $42a^3b^9c^7$

C. $-42a^4b^9c^6$

D. $42a^4b^9c^6$

28. If two complementary angles have measures of $2x + 21$ and $3x - 26$, the smaller angle has a measure of ____.

A. 19

B. 31

C. 38

D. 59

29. Two supplementary angles are in the ratio $2:3$. Find the smaller angle of the two.

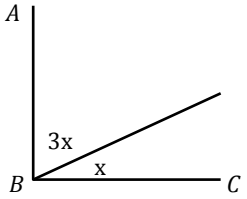
A. 72°

B. 36°

C. 108°

D. 30°

30. If $\overline{AB} \perp \overline{BC}$, then x must equal



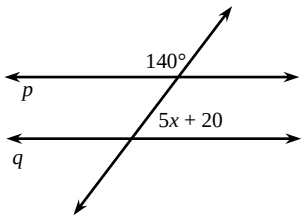
A. 22.5

B. 4

C. 25

D. 45

31. If $p//q$, then x must equal



E. 3

F. 4

G. 5

H. 6

32. Which of the following are sides of a right triangle?

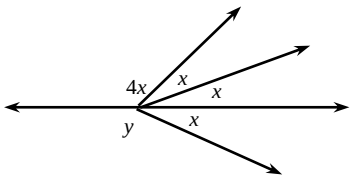
A. 2, 5, 8

B. 1, 2, 3

C. 5, 12, 13

D. 3, 7, 7

33. In the figure below, find the value of y.



A. 160°

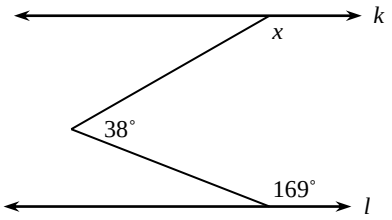
B. 150°

C. 125°

D. 120°

34. Which of the following is not a function?
A. $\{(-1, 3), (-2, 1), (-3, 3), (-6, 4)\}$
B. $\{(2, 1), (2, 2), (2, 3), (2, 4)\}$
C. $\{(2, \pi), (-1, \pi), (0, \pi), (1, \pi)\}$
D. $\{(-2, -2), (0, 0), (2, 2), (4, 4)\}$

35. Refer to the figure. If $k \parallel l$, then find the measure of angle x.



A. 153°
B. 142°
C. 97°
D. 11°

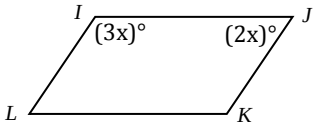
36. $\dfrac{a^3b^{-6}c^4\,d^{-3}}{a^{-7}b^4c^{-6}d^{-3}} =$
A. $\dfrac{a^4c^{10}}{b^{10}}$
B. $\dfrac{a^{10}c^{10}}{b^{10}}$
C. $\dfrac{a^{10}b^{10}}{c^{10}}$
D. $\dfrac{b^4c^{10}}{a^{10}}$

37. The equation of the line that passes through the points $(-3, -1)$ and $(-5, 2)$.
A. $3x + 2y - 11 = 0$
B. $3x + 2y + 11 = 0$
C. $3x - 2y + 7 = 0$
D. $3x + 2y - 7 = 0$

38. Find the exact value of $8\sin 210° - 4\cos 300°$.
A. -12
B. -6
C. 0
D. 2

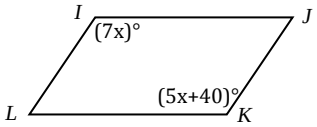
39. Find the equation of the line that passes through $(2, -3)$ and parallel to $3x + 4y = 7$.
A. $3x + 4y = -6$
B. $3x - 4y = 18$
C. $3x + 4y = 6$
D. $3x - 4y = 10$

40. If $IJKL$ is a parallelogram, find the value of x.



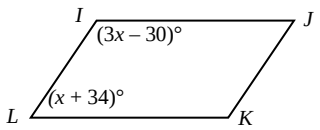
A. 36
B. 45
C. 102
D. 132

41. If $IJKL$ is a parallelogram, find the value of x.



A. 20
B. 30
C. 45
D. 60

42. If $IJKL$ is a parallelogram, then $m\angle K$ = ____.



A. 44°

B. 78°

C. 102°

D. 132°

43. $\left(\dfrac{15x^{-2}y^4z^6}{3x^4y^{-3}z^5}\right)^3$

A. $\dfrac{125y^{21}z^3}{x^{18}}$

B. $\dfrac{5y^{21}z^3}{x^{18}}$

C. $\dfrac{125y^{10}z^4}{x^9}$

D. $\dfrac{5y^{10}z^4}{x^9}$

44. Find the midpoint of (–2, 6) and (–3, 8).

A. (–1, –7)

B. (–5, 14)

C. (–2.5, 7)

D. (–2.5, 5)

45. If (2, 0) is the midpoint of (–2, –2) and (x, y), then (x, y) = _____.

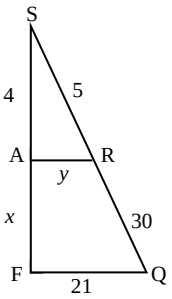
A. (0, –2)

B. (6, 2)

C. (–2, 0)

D. (4, 6)

46. Find the values of x and y.



A. $x = 21$ and $y = 3$

B. $x = 24$ and $y = 3$

C. $x = 24$ and $y = 4$

D. $x = 21$ and $y = 4$

47. Point M is between C and D. If the length of $\overline{MC}$ is 3 units more than five times the length of $\overline{MD}$ and CD = 39 units, find the length of $\overline{MC}$.

A. 6

B. 18

C. 33

D. 42

48. Point M is between A and B. If $AM = 3x + 7$, $MB = 7x - 3$, and $AB = 24$, find the value of x.

A. 6

B. 8

C. 2

D. 5

49. Point M is the midpoint of AB. If $AM = 3x + 8$ and $MB = 7x - 12$, find the value of x.

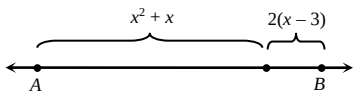
A. 6

B. 8

C. 2

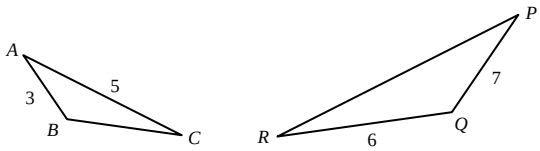
D. 5

50. Refer to the figure below. If AB = 48 units, find the value of x.



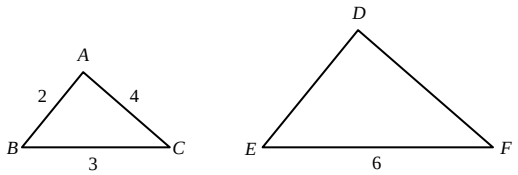
A. 1
B. 3
C. 6
D. 9

51. $\Delta ABC \sim \Delta PQR$. Find PR.



A. $\frac{35}{3}$
B. $\frac{21}{5}$
C. 10
D. 14

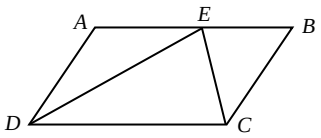
52. $\Delta ABC \sim \Delta DEF$. Find DE.



A. 3
B. 4
C. 5
D. 6

53. The difference of two numbers is 3. The smaller number is one more than two-thirds of the larger. Find the larger number.
A. 5
B. 9
C. 12
D. 15

54. $ABCD$ is a parallelogram. The angle bisectors of $\angle D$ and $\angle C$ meet at E. Find the measure of $\angle DEC$.



A. 60°
B. 80°
C. 90°
D. 100°

55. In a certain class, the number of boys to girls is in the ratio 3:5. How many students are there in the class if the number of girls is six less than twice the number of boys?
A. 48
B. 36
C. 30
D. 6

56. A farmer has enough grain to feed 50 cattle for 10 days. He sells 10 cattle. For how many days will the grain last now?
A. 12 ½
B. 15
C. 12
D. 8

57. Lora is now twice as old as Krista. Four years ago Lora was three times as old as Krista. How old is Lora?
A. 12
B. 15
C. 16
D. 20

58. It takes 9 hours for 4 workers to finish a certain job. How long will it take 6 workers to finish the same job?
A. 5 hours
B. $2\frac{2}{3}$ hours
C. $8\frac{3}{4}$ hours
D. 6 hours

59. Suppose one painter can paint the entire house in twelve hours, and the second painter takes eight hours. How long would it take the two painters together to paint the house?
A. $5\frac{1}{2}$ hours
B. $4\frac{4}{5}$ hours
C. $4\frac{1}{5}$ hours
D. 4 hours

60. Solve for x: $7x^2 - 5x = 2$
A. $1, -\frac{2}{7}$
B. $-1, \frac{2}{7}$
C. $2, -\frac{1}{7}$
D. $-2, \frac{1}{7}$